
N100 Financial Intelligence Platform

A Production-Grade Financial Intelligence and Business Analytics Platform for Nifty 100 Companies

PROJECT REPORT

Submitted by

Arin Bhardwaj

M.Sc. Mathematics and Computing
National Institute of Technology Warangal

Academic Year: 2025-2027

Technologies Used: Python, Pandas, NumPy, SQLite, Plotly, Streamlit, Git, GitHub

GitHub Repository:

https://github.com/Bhardwaj-Arin/N100_FINANCIAL_INTELLIGENCE_PLATFORM

Live Application:

<https://n100financialintelligenceplatform-njqenrfgue6evxbdjcmj.streamlit.app/>

Submission Date: July 2026

Declaration

I hereby declare that the project titled "N100 Financial Intelligence Platform" is an original work carried out by me as part of my academic learning and personal portfolio development. This project has been designed, implemented, tested, documented, and deployed using modern data science, financial analytics, and visualization technologies.

The work presented in this report has been developed independently using publicly available financial datasets, open-source Python libraries, and standard software development practices. Every effort has been made to ensure the accuracy, reliability, and authenticity of the analyses, dashboards, and documentation included in this report.

This project has not been submitted, either in whole or in part, for the award of any other degree, diploma, or certification at any university or institution. All external resources, libraries, frameworks, and references used during the development of this project have been appropriately acknowledged in the References section of this report.

The N100 Financial Intelligence Platform demonstrates my understanding of data engineering, financial data analysis, exploratory data analysis, feature engineering, relational databases, interactive dashboard development, business intelligence, and deployment using Streamlit. The project reflects my practical skills in building an end-to-end analytics application capable of transforming raw financial data into meaningful business insights.

I take full responsibility for the originality and authenticity of the work presented in this report.

Submitted by,

Arin Bhardwaj

M.Sc. Mathematics & Computing

National Institute of Technology Warangal

Date: _____

Signature: _____

Acknowledgement

I would like to express my sincere gratitude to everyone who directly or indirectly contributed to the successful completion of this project, "N100 Financial Intelligence Platform." The development of this project has been a valuable learning experience that strengthened my understanding of data science, financial analytics, software development, and business intelligence.

I am grateful to the faculty members of the Department of Mathematics, National Institute of Technology Warangal, for providing a strong academic foundation that encouraged analytical thinking and problem-solving. Their guidance throughout my academic journey has played an important role in enhancing my technical knowledge.

I would also like to acknowledge the developers and maintainers of the open-source technologies and libraries used in this project, including Python, Pandas, NumPy, SQLite, Plotly, Streamlit, and GitHub. These tools made it possible to build a production-style financial analytics platform while following modern software engineering practices.

A special thanks goes to the online learning communities, technical documentation, and educational resources that helped me understand financial metrics, dashboard development, database management, and interactive visualization techniques. Their comprehensive tutorials and documentation significantly contributed to overcoming technical challenges encountered during the project.

Finally, I would like to thank my family and friends for their constant encouragement, motivation, and support throughout the development of this project. Their confidence in my abilities inspired me to complete this work with dedication and enthusiasm.

I hope this project demonstrates my practical skills in financial data analytics, data engineering, and interactive dashboard development while serving as a strong foundation for future research and professional opportunities.

Table of Contents

Page	Section
1	Cover Page
2	Declaration
3	Acknowledgement
4	Table of Contents
5	Executive Summary
6	Problem Statement
7	Project Objectives
8	Technology Stack
9	System Architecture
10	Dataset Description
11	Data Pipeline
12	Feature Engineering
13-20	Dashboard Modules
21-24	Cash Flow Intelligence
25-28	Portfolio Analytics
29	Deployment
30	Testing and Validation
31	Results and Achievements
32	Conclusion and Future Scope
33	References and Repository Assets

Note: The GitHub repository and Streamlit application links are embedded as active hyperlinks in this PDF.

Executive Summary

The N100 Financial Intelligence Platform is a production-style financial analytics application developed to provide a centralized and interactive environment for analyzing companies listed in the Nifty 100 index. The project integrates financial data processing, business intelligence, visualization, and decision-support tools into a single web-based platform, enabling users to evaluate company performance through an intuitive interface.

Traditional financial analysis often requires users to navigate multiple reports, spreadsheets, and financial statements to understand a company's performance. This process is time-consuming and makes comparisons across companies and sectors difficult. The N100 Financial Intelligence Platform addresses this challenge by consolidating financial information into an integrated dashboard that supports data-driven decision-making.

The application follows a complete analytics pipeline beginning with data collection and validation, followed by preprocessing, feature engineering, financial metric calculation, database integration, and interactive visualization. Using Python, Pandas, SQLite, Plotly, and Streamlit, the platform transforms raw financial data into meaningful insights through dynamic dashboards and analytical reports.

The platform includes ten major analytical modules: Dashboard, Company Profile, Peer Comparison, Sector Analysis, Rankings, Investment Screener, Trend Analysis, NLP Insights, Cash Flow Intelligence, and Portfolio Analytics. These modules allow users to compare companies, analyze sector performance, monitor financial health, evaluate cash flow quality, screen investment opportunities, and construct diversified portfolios using interactive visualizations.

Overall, this project demonstrates practical skills in data engineering, financial analytics, software development, dashboard design, database management, and deployment. It represents an end-to-end business intelligence solution capable of supporting investors, students, researchers, and financial analysts in making informed decisions based on structured financial data.

Figure 5.1: Overall Dashboard of the N100 Financial Intelligence Platform.

Project Metric	Value
Processed records	6,972
Companies in latest snapshot	92
Broad sectors	10
Analytical fields	94
Average Financial Health Score	67.80
Highest Financial Health Score	88.33

Problem Statement

Financial analysis plays a critical role in investment decisions, corporate strategy, and business evaluation. However, most publicly available financial information is distributed across multiple reports, websites, and spreadsheets, making it difficult for users to obtain a unified view of a company's financial performance. Analysts often spend significant time collecting, cleaning, and comparing data before meaningful insights can be generated.

For investors, comparing companies across different sectors requires evaluating numerous financial metrics such as profitability, return ratios, cash flow, growth, and financial stability. Performing these comparisons manually is inefficient and increases the likelihood of errors or inconsistent analysis. Similarly, students and researchers frequently lack an integrated platform that combines financial data, visual analytics, and business intelligence tools in a single environment.

The absence of centralized financial intelligence also limits portfolio evaluation and sector-level benchmarking. Without interactive dashboards, users struggle to identify market leaders, compare peer companies, analyze historical trends, or screen companies using customizable financial criteria.

The N100 Financial Intelligence Platform was developed to address these challenges by creating an end-to-end analytics solution for companies in the Nifty 100 index. The platform automates data processing, feature engineering, financial metric computation, ranking, visualization, and portfolio analysis. It enables users to explore financial information through interactive dashboards rather than static spreadsheets, making analysis faster, more intuitive, and more reliable.

By integrating multiple analytical modules into a single Streamlit application, the project demonstrates how modern data science and business intelligence techniques can simplify financial analysis and support informed decision-making for investors, researchers, and students.

Project Objectives

The primary objective of the N100 Financial Intelligence Platform is to design and develop a comprehensive financial analytics platform that transforms raw financial statement data into meaningful, interactive, and actionable business insights. The platform is intended to simplify financial analysis by integrating data processing, visualization, and decision-support tools into a single application.

Specific Objectives

1. Develop a centralized platform for analyzing financial information of Nifty 100 companies.
2. Build an automated data processing pipeline for cleaning, validating, and transforming financial datasets.
3. Calculate important financial metrics and business KPIs that assist in evaluating company performance.
4. Design interactive dashboards to visualize profitability, growth, cash flow, and financial health.
5. Enable peer comparison and sector-wise benchmarking through dynamic visualizations.
6. Rank companies using standardized financial indicators and a Financial Health Score.
7. Provide an investment screener that allows users to filter companies based on customizable financial criteria.
8. Analyze historical financial trends to identify long-term business performance patterns.
9. Present rule-based NLP insights that summarize key financial observations in an easily understandable format.
10. Support portfolio analysis by comparing multiple companies, evaluating diversification, and identifying top-performing investments.
11. Demonstrate an end-to-end data analytics workflow, including data engineering, database integration, dashboard development, testing, and deployment.

Expected Outcomes

By achieving these objectives, the platform provides users with an efficient environment for exploring financial data, comparing companies, identifying investment opportunities, and making informed decisions using interactive business intelligence tools.

Technology Stack

The N100 Financial Intelligence Platform was developed using a modern data science and software development stack that supports data processing, financial analysis, interactive visualization, database management, and web application deployment. Each technology was selected to address a specific requirement within the project workflow, resulting in a scalable and maintainable analytics platform.

Programming Language

Python served as the primary programming language for implementing the complete application. Its rich ecosystem of data science libraries enabled efficient data processing, feature engineering, visualization, and dashboard development.

Data Processing

Pandas was used for data cleaning, transformation, aggregation, feature engineering, and manipulation of financial datasets. NumPy supported numerical computations and optimized mathematical operations on large datasets.

Database

SQLite was used as the backend database to store processed financial data. The lightweight relational database simplified data retrieval while ensuring consistency and efficient querying throughout the application.

Data Visualization

Plotly was used to build highly interactive visualizations including bar charts, line charts, scatter plots, treemaps, box plots, pie charts, and financial comparison dashboards. These visualizations improve the interpretability of complex financial information.

Web Application

Streamlit was used to develop the interactive web application. It enabled rapid development of user-friendly dashboards where users can explore company profiles, rankings, financial metrics, cash flow intelligence, and portfolio analytics through an intuitive interface.

Version Control

Git and GitHub were used for version control, collaborative development, project documentation, and source code management throughout the development lifecycle.

Development Environment

Visual Studio Code (VS Code) was used as the primary integrated development environment (IDE) for writing, debugging, and maintaining the project's source code.

Technology Summary

- **Programming Language: Python**
- **Data Processing: Pandas, NumPy**
- **Database: SQLite**
- **Visualization: Plotly**
- **Web Framework: Streamlit**
- **Version Control: Git & GitHub**
- **IDE: Visual Studio Code**

Figure 8.1: Technology Stack Used in the N100 Financial Intelligence Platform.

System Architecture

The N100 Financial Intelligence Platform follows a layered architecture that separates data ingestion, preprocessing, feature engineering, scoring, storage, and visualization. This separation improves maintainability and supports future enhancements such as live data feeds, model services, and automated report generation.

Layered System Architecture

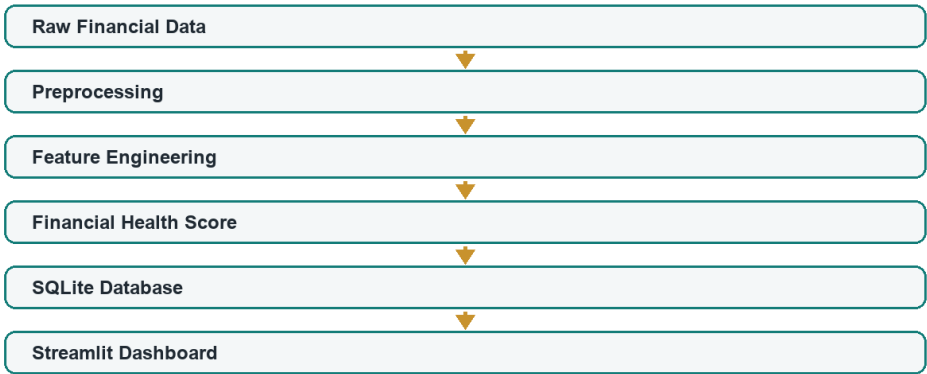


Figure 9.1: Layered architecture of the N100 Financial Intelligence Platform.

Dataset Description

The N100 Financial Intelligence Platform is built upon a structured financial dataset containing information for companies included in the Nifty 100 index. The dataset serves as the foundation for all analytical modules, enabling financial comparison, ranking, trend analysis, and portfolio evaluation.

The dataset contains company-level financial information collected from publicly available financial reports and organized into a tabular format suitable for analysis. Each record represents a company and includes multiple financial attributes covering profitability, valuation, cash flow, growth, efficiency, and balance sheet performance. These variables are further processed to generate business insights and interactive visualizations.

Prior to analysis, the dataset undergoes validation and preprocessing to ensure consistency and accuracy. Missing values, duplicate records, inconsistent formats, and invalid entries are identified and handled appropriately. This preprocessing stage improves data quality and ensures reliable downstream analytics.

Major Categories of Features

- Company Information
 - Company Name
 - Sector
 - Industry
- Profitability Metrics
 - Revenue
 - Net Profit
 - EBITDA
 - Operating Margin
 - Net Profit Margin
- Return Ratios
 - Return on Equity (ROE)
 - Return on Capital Employed (ROCE)
 - Return on Assets (ROA)
- Valuation Metrics
 - Market Capitalization
 - Price-to-Earnings (P/E) Ratio
 - Price-to-Book (P/B) Ratio
- Cash Flow Metrics
 - Operating Cash Flow
 - Free Cash Flow
 - Capital Expenditure
- Growth Indicators
 - Revenue Growth
 - Profit Growth
 - Earnings Growth

Dataset Usage

The processed dataset supports multiple analytical tasks, including company profiling, peer comparison, sector benchmarking, financial rankings, investment screening, historical trend analysis, cash flow intelligence, NLP-based insights, and portfolio analytics. The engineered dataset acts as the primary source for the SQLite database used by the Streamlit application.

Figure 10.1: Overview of the Financial Dataset Used in the N100 Financial Intelligence Platform.

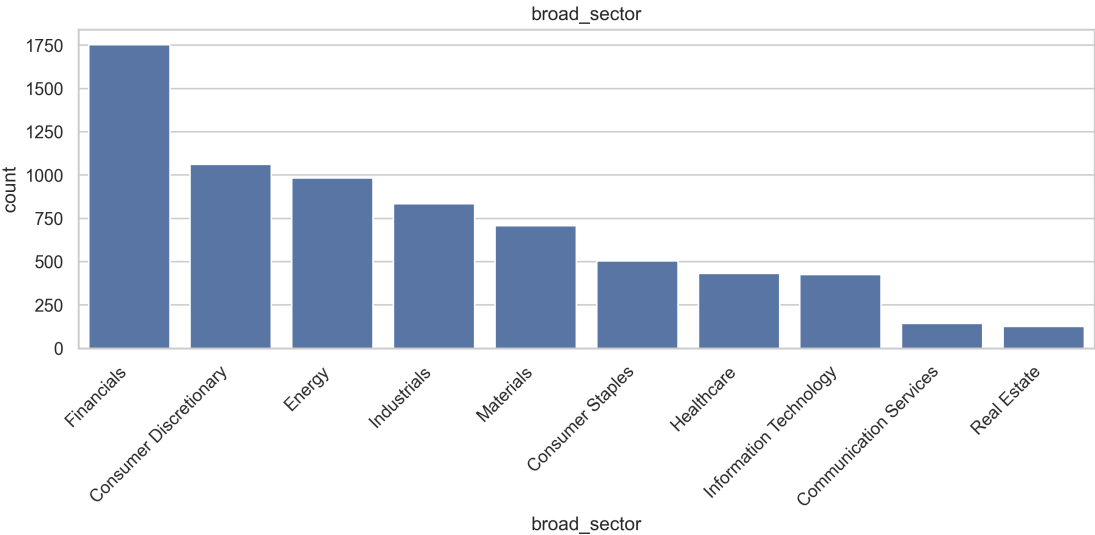


Figure 10.1: Analytical visualization from the project reports.

Data Pipeline

The data pipeline converts raw financial inputs into a standardized analytical dataset. Raw Excel files are loaded, validated, cleaned, joined, and transformed into the master feature table used by the dashboard and scoring engine.

Pipeline Stage	Purpose
Data ingestion	Load company, balance sheet, cash flow, ratios, market, and sector data.
Validation	Check required columns, missing values, duplicates, and data types.
Preprocessing	Standardize names, numerical fields, categories, and join keys.
Feature engineering	Create profitability, stability, valuation, cash-flow, and sector signals.
Storage	Persist processed records and score outputs in CSV and SQLite.
Dashboard delivery	Serve interactive Streamlit pages through reusable loading functions.

Project Metric	Value
Processed records	6,972
Companies in latest snapshot	92
Broad sectors	10
Analytical fields	94
Average Financial Health Score	67.80
Highest Financial Health Score	88.33

Feature Engineering

Feature engineering is one of the most important stages in the N100 Financial Intelligence Platform because it transforms raw financial data into meaningful business indicators that support decision-making. Rather than relying solely on values directly available in financial statements, the platform derives additional metrics that provide a more comprehensive view of a company's financial performance.

After data cleaning and validation, several financial features are computed using the original dataset. These engineered features simplify company comparison, improve dashboard visualization, and enable ranking and screening of companies based on standardized financial indicators.

Financial Health Score

A composite Financial Health Score is generated by combining multiple financial metrics into a single performance indicator. This score provides a quick overview of a company's overall financial strength and is extensively used throughout the dashboard for benchmarking and rankings.

Profitability Indicators

Key profitability metrics such as Operating Margin, Net Profit Margin, EBITDA Margin, Return on Equity (ROE), Return on Capital Employed (ROCE), and Return on Assets (ROA) are utilized to evaluate operational efficiency and shareholder returns. These indicators allow users to compare companies across different industries using standardized measures.

Cash Flow Metrics

Cash flow-related features, including Operating Cash Flow, Free Cash Flow, Capital Expenditure, and Cash Conversion indicators, are incorporated to evaluate liquidity, sustainability, and the quality of earnings. These metrics play a central role in the Cash Flow Intelligence module.

Growth Indicators

Growth-related features such as Revenue Growth, Profit Growth, and Earnings Growth are included to measure business expansion over time. These indicators help identify companies demonstrating consistent long-term performance.

Ranking Variables

The engineered dataset also contains ranking variables that support the Rankings module and Investment Screener. Companies can be ranked based on profitability, financial health, cash flow performance, and other key financial measures, enabling users to identify top-performing organizations efficiently.

Business Value

The engineered features significantly improve the analytical capability of the platform by converting complex financial information into actionable insights. They serve as the foundation for company benchmarking, portfolio analytics, sector comparison, trend analysis, and financial decision support throughout the application.

Figure 12.1: Feature Engineering Workflow Used in the N100 Financial Intelligence Platform.

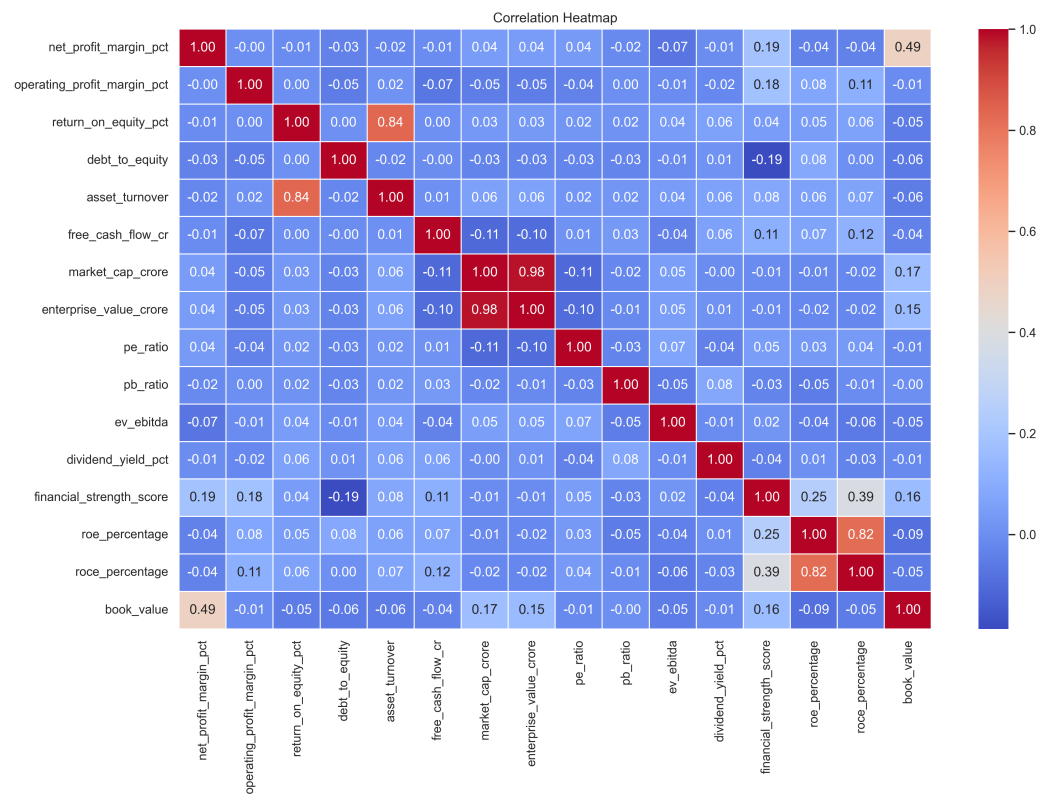


Figure 12.1: Analytical visualization from the project reports.

Dashboard Overview

The Dashboard serves as the central landing page of the N100 Financial Intelligence Platform and provides users with a high-level overview of the financial ecosystem of the Nifty 100 companies. It is designed to present the most important financial indicators, summary statistics, and visual insights in a single, interactive interface, enabling users to quickly understand the overall market landscape before exploring individual analytical modules.

The dashboard follows a business intelligence approach by combining key performance indicators (KPIs), interactive charts, filters, and summary tables into a unified layout. Users can instantly access important financial information without manually navigating multiple datasets or reports.

Key Features

- Interactive KPI cards displaying important financial metrics.
- Company and sector selection for customized analysis.
- Dynamic visualizations built using Plotly.
- Financial Health Score overview.
- Summary statistics for quick business assessment.
- Interactive charts supporting exploratory financial analysis.
- Responsive layout designed for ease of use.

Business Benefits

The dashboard significantly reduces the effort required to analyze financial data by consolidating essential information into a single screen. It serves as the primary navigation point for the application and provides immediate visibility into company performance, sector trends, and overall financial health.

The dashboard is designed to support multiple user groups, including investors, students, researchers, financial analysts, and business professionals. Through intuitive visualizations and interactive components, users can quickly identify trends, compare organizations, and decide which analytical module to explore next.

Overall, the Dashboard acts as the executive summary of the entire platform, offering an efficient starting point for comprehensive financial analysis.

Figure 13.1: Main Dashboard of the N100 Financial Intelligence Platform.

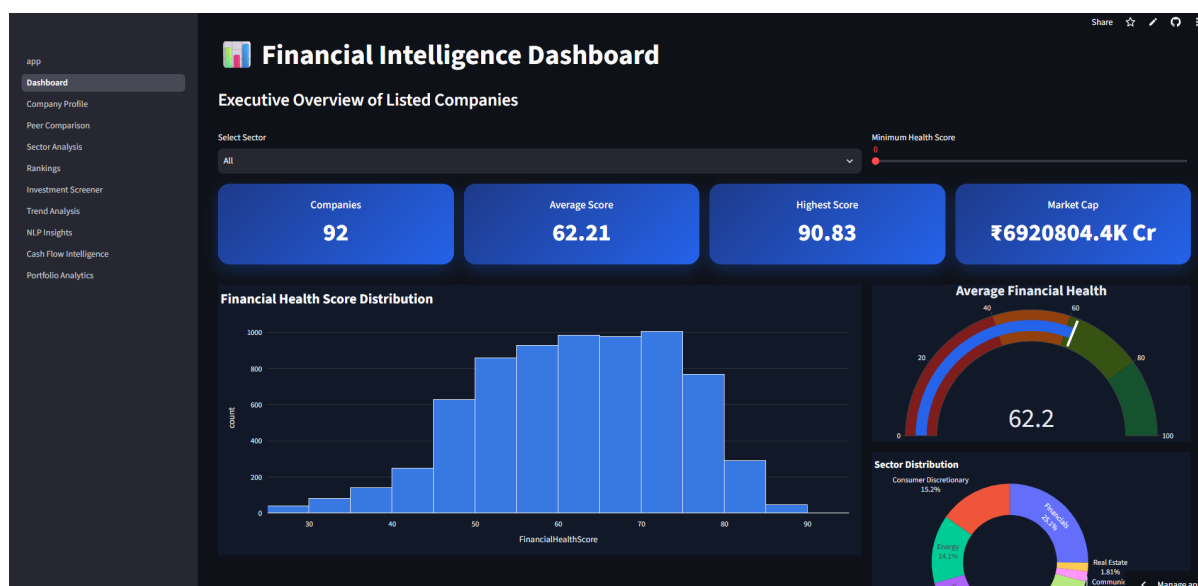


Figure 13.1: Application screenshot supporting this module.

Company Profile

The Company Profile module provides a comprehensive overview of an individual company’s financial performance. It serves as a centralized dashboard where users can explore detailed financial information, business metrics, and performance indicators for a selected company. This module is designed to simplify company-level analysis by presenting complex financial data through intuitive visualizations and interactive components.

Users can select any company from the Nifty 100 universe and instantly view its financial profile. The module consolidates important financial metrics, profitability indicators, valuation measures, return ratios, cash flow statistics, and business performance into a single interface, eliminating the need to manually examine multiple financial statements.

Key Features

- Interactive company selection.
- Overview of important financial KPIs.
- Profitability metrics such as ROE, ROCE, and profit margins.
- Revenue and earnings analysis.
- Financial Health Score for overall business assessment.
- Interactive charts illustrating financial performance.
- Easy-to-read summary tables and visual indicators.

Business Value

The Company Profile module enables users to perform a detailed evaluation of an organization’s financial condition before making investment or business decisions. Investors can quickly assess profitability, operational efficiency, and financial stability, while students and researchers can use the module to understand the relationship between different financial metrics.

By presenting financial information in a structured and visually appealing format, the module improves decision-making efficiency and reduces the complexity associated with traditional financial statement analysis.

Overall, the Company Profile acts as the foundation for company-specific analysis within the N100 Financial Intelligence Platform and supports deeper exploration through the remaining analytical modules.

Figure 14.1: Company Profile Dashboard of the N100 Financial Intelligence Platform.

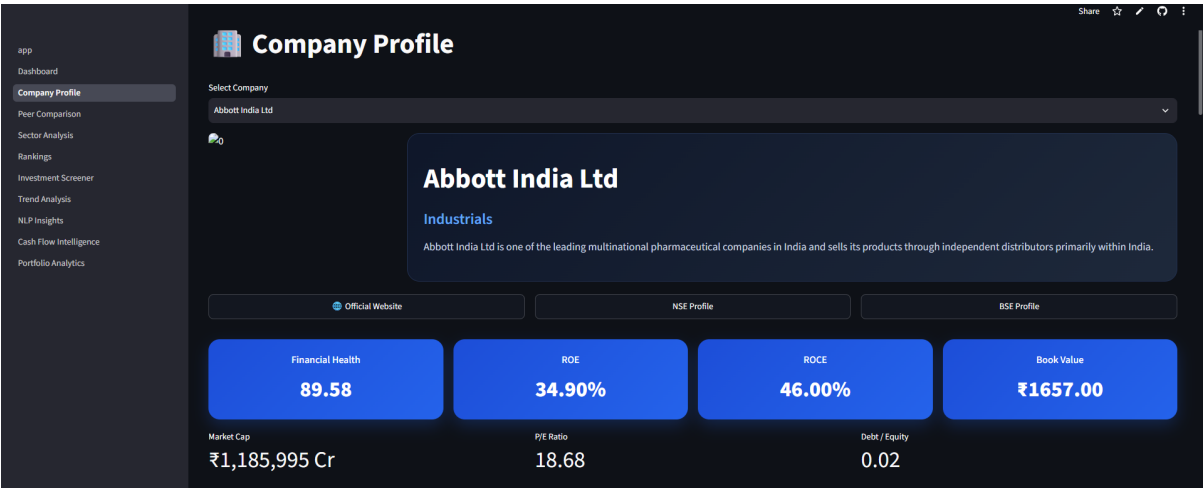


Figure 14.1: Application screenshot supporting this module.

Peer Comparison

The Peer Comparison module enables users to evaluate the financial performance of a selected company against its industry peers. Rather than analyzing organizations in isolation, this module provides a comparative perspective by displaying multiple companies side by side using standardized financial metrics and interactive visualizations.

Users can select a company and compare it with other companies operating in the same sector or industry. The module presents key financial indicators including profitability, return ratios, valuation measures, growth metrics, cash flow indicators, and the Financial Health Score. This comparison helps users understand where a company stands relative to its competitors and identify both strengths and areas for improvement.

Key Features

- Side-by-side comparison of multiple companies.
- Sector-specific benchmarking.
- Interactive charts and comparison tables.
- Financial Health Score comparison.
- Analysis of profitability, valuation, growth, and return ratios.
- Dynamic filtering for customized comparisons.

Business Value

Peer comparison is an essential component of financial analysis because absolute financial values often provide limited insight without proper context. By comparing companies operating within the same sector, users can identify market leaders, evaluate competitive positioning, and understand industry standards.

This module is particularly valuable for investors seeking investment opportunities, researchers performing comparative studies, and students learning financial statement analysis. Interactive visualizations simplify complex comparisons and allow users to identify trends and outliers more efficiently than traditional spreadsheet-based analysis.

Overall, the Peer Comparison module strengthens the analytical capability of the N100 Financial Intelligence Platform by providing an objective framework for benchmarking company performance against comparable organizations.

Figure 15.1: Peer Comparison Dashboard of the N100 Financial Intelligence Platform.

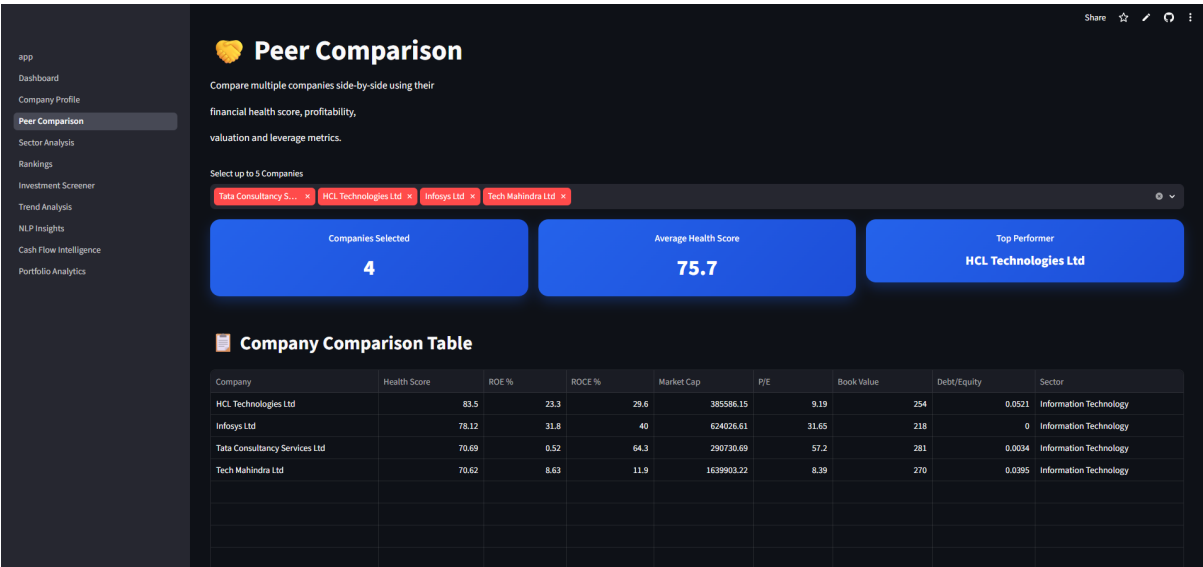


Figure 15.1: Application screenshot supporting this module.

Sector Analysis

The Sector Analysis module provides a comprehensive view of financial performance across different sectors represented within the Nifty 100 index. Instead of focusing on individual companies, this module aggregates financial information at the sector level, allowing users to identify industry trends, compare sector performance, and evaluate the overall strength of different segments of the market.

The module organizes companies according to their respective sectors and computes summary statistics using important financial indicators. Through interactive charts and visual dashboards, users can compare sectors based on profitability, return ratios, market performance, financial health, and cash flow characteristics. This approach provides a broader understanding of market dynamics and highlights which industries are outperforming or underperforming.

Key Features

- Sector-wise financial comparison.
- Interactive sector selection and filtering.
- Comparison of average Financial Health Scores.
- Profitability and return ratio analysis across sectors.
- Interactive bar charts, treemaps, and distribution plots.
- Ranking of sectors based on financial performance.
- Summary tables containing important business metrics.

Business Value

Sector-level analysis is essential because companies operate under different market conditions, regulatory environments, and business models. Comparing organizations only at the company level may overlook broader industry trends. By evaluating financial indicators collectively, users gain insight into which sectors demonstrate stronger profitability, stability, and long-term growth potential.

For investors, this module assists in identifying attractive industries for diversification and capital allocation. Researchers can study sector-specific financial behavior, while students can better understand how macroeconomic conditions influence different industries. The interactive visualizations simplify complex comparisons and support faster, data-driven decision making.

Overall, the Sector Analysis module enhances the platform by providing a market-wide perspective that complements company-level analysis and enables users to evaluate financial performance across the entire Nifty 100 ecosystem.

Figure 16.1: Sector Analysis Dashboard of the N100 Financial Intelligence Platform.

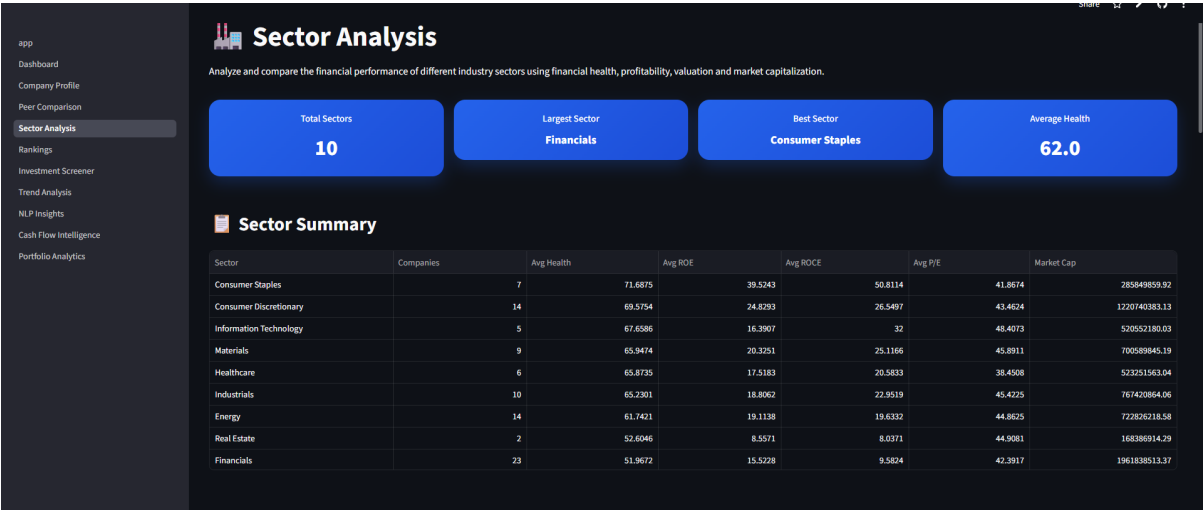


Figure 16.1: Application screenshot supporting this module.

Rankings

The Rankings module provides a systematic approach to evaluating and ranking companies based on their financial performance. It transforms multiple financial indicators into easy-to-understand leaderboards, allowing users to identify the strongest and weakest companies within the Nifty 100 index. By presenting companies in ranked order, the module simplifies financial comparison and supports objective investment analysis.

The ranking process is based on key financial metrics such as profitability, return ratios, cash flow indicators, valuation measures, and the Financial Health Score. These metrics are calculated during the feature engineering stage and standardized to ensure meaningful comparisons across companies from different sectors.

Users can dynamically sort companies according to various financial indicators, making it easy to identify market leaders, compare business performance, and evaluate organizations against their peers. Interactive filters allow users to customize rankings based on specific sectors or financial metrics, providing greater flexibility during analysis.

Key Features

- Company rankings based on Financial Health Score.
- Profitability-based rankings using financial ratios.
- Dynamic sorting and filtering capabilities.
- Sector-wise ranking comparison.
- Interactive tables and visual leaderboards.
- Easy identification of top and bottom performers.
- Support for comparative investment analysis.

Business Value

Ranking companies based on standardized financial indicators enables faster and more objective decision-making than manually reviewing financial statements. Investors can quickly identify financially strong organizations, while analysts can evaluate relative performance across sectors. Students and researchers can also use the rankings to understand how financial metrics influence company performance and market positioning.

The Rankings module plays an important role in the overall platform by converting complex financial information into intuitive leaderboards that are easy to interpret. It complements the Company Profile and Peer Comparison modules by providing a broader market-wide perspective on financial performance.

Overall, this module improves the efficiency of financial analysis by enabling users to rapidly discover high-performing companies and evaluate their competitive standing within the Nifty 100 ecosystem.

Figure 17.1: Rankings Dashboard of the N100 Financial Intelligence Platform.

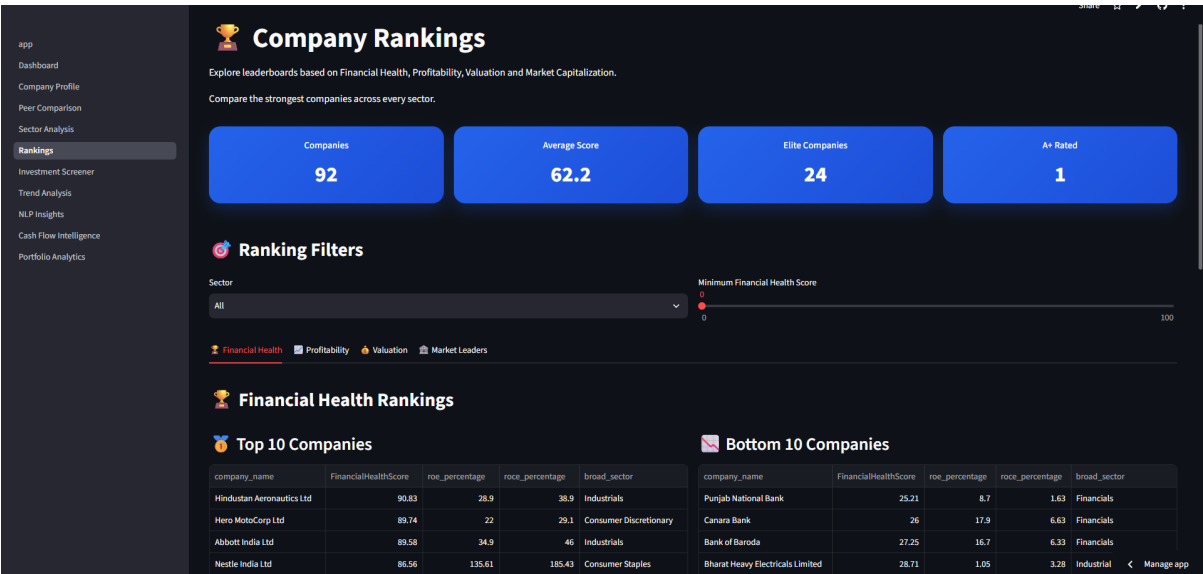


Figure 17.1: Application screenshot supporting this module.

Investment Screener

The Investment Screener module enables users to identify companies that satisfy specific financial criteria through an interactive filtering system. Instead of manually examining financial statements for every company, users can apply multiple screening conditions to quickly discover organizations that align with their investment objectives.

The screener operates on the processed financial dataset and allows filtering based on important business indicators such as Financial Health Score, profitability ratios, return metrics, valuation measures, growth indicators, and cash flow characteristics. By combining multiple filters, users can narrow the list of companies and focus only on those that meet their predefined requirements.

This module is particularly useful for investors who follow value investing, growth investing, or quality investing strategies. Researchers and students can also use the screener to study how different financial parameters influence company selection and investment decisions.

Key Features

- Interactive filtering based on multiple financial metrics.
- Sector-wise company screening.
- Financial Health Score filters.
- Profitability and return ratio screening.
- Valuation-based company selection.
- Dynamic tables displaying filtered results.
- Real-time updates as filter values change.

Business Value

Investment decisions often require evaluating hundreds of companies using numerous financial indicators. Performing this task manually is inefficient and prone to errors. The Investment Screener simplifies this process by automatically identifying companies that satisfy user-defined conditions, significantly reducing analysis time while improving decision quality.

The module also encourages objective decision-making by relying on standardized financial metrics rather than subjective judgments. Interactive filtering helps users explore different investment scenarios and compare companies under various financial conditions.

Overall, the Investment Screener transforms the N100 Financial Intelligence Platform into a practical decision-support tool by enabling fast, flexible, and data-driven company selection for investment analysis.

Figure 18.1: Investment Screener Dashboard of the N100 Financial Intelligence Platform.



Figure 18.1: Application screenshot supporting this module.

Trend Analysis

The Trend Analysis module provides a historical perspective on the financial performance of companies by examining how key financial metrics change over time. While point-in-time financial values offer a snapshot of business performance, trend analysis reveals long-term patterns, consistency, and growth trajectories that are essential for making informed investment and strategic decisions.

This module enables users to analyze historical changes in profitability, revenue, earnings, return ratios, cash flow, and the Financial Health Score using interactive time-series visualizations. By observing these trends, users can determine whether a company is improving, remaining stable, or experiencing financial deterioration over multiple reporting periods.

Interactive charts allow users to compare different metrics simultaneously, making it easier to identify relationships between growth, profitability, and operational performance. The dynamic filtering options also enable company-specific trend analysis, allowing users to focus on organizations of interest without unnecessary distractions.

Key Features

- Interactive time-series visualizations.
- Historical analysis of financial metrics.
- Revenue and profit trend monitoring.
- Financial Health Score progression.
- Comparison of multiple financial indicators over time.
- Dynamic company selection and filtering.
- Easy interpretation through interactive Plotly charts.

Business Value

Historical analysis plays a crucial role in financial decision-making because investment decisions should not rely solely on current financial performance. Companies demonstrating stable and consistent growth often provide stronger long-term investment opportunities than organizations with volatile financial behavior.

The Trend Analysis module enables investors to identify sustainable business growth, evaluate financial consistency, and recognize significant changes in operational performance. Researchers and students can also use historical visualizations to understand how financial indicators evolve over time and how different events influence business performance.

Overall, this module enhances the analytical capabilities of the N100 Financial Intelligence Platform by transforming historical financial data into meaningful visual insights that support long-term strategic analysis.

Figure 19.1: Trend Analysis Dashboard of the N100 Financial Intelligence Platform.

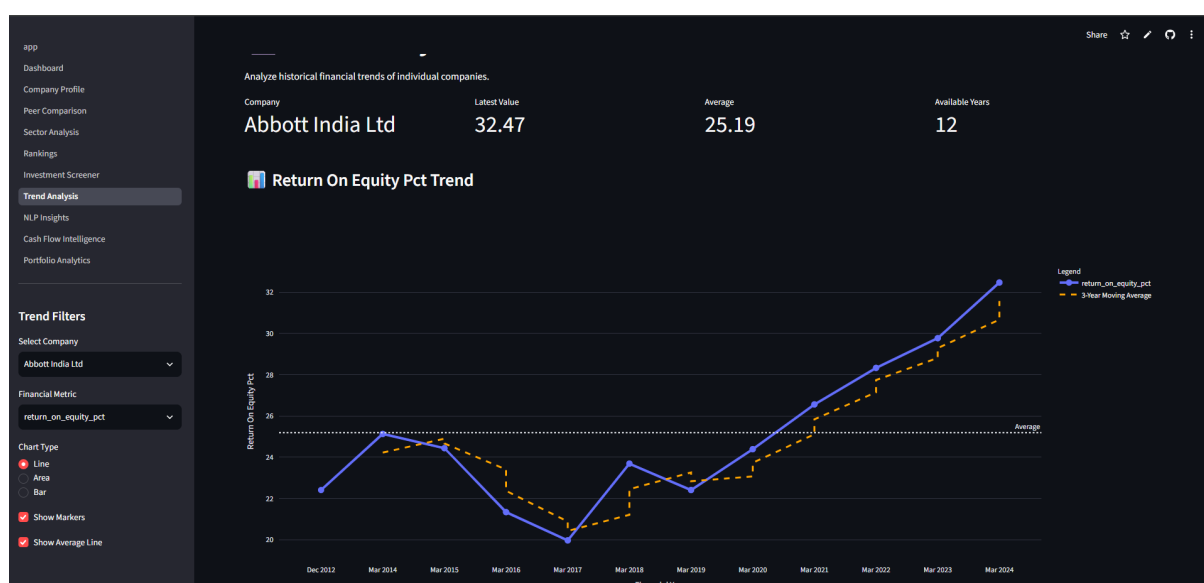


Figure 19.1: Application screenshot supporting this module.

NLP Insights

The NLP Insights module enhances the N100 Financial Intelligence Platform by transforming numerical financial data into concise, human-readable business observations. Instead of requiring users to interpret multiple financial ratios and charts manually, this module generates meaningful textual insights that summarize the financial condition and performance of a selected company.

The implementation follows a rule-based Natural Language Processing (NLP) approach. Financial metrics calculated during the feature engineering stage are evaluated against predefined business rules to generate descriptive insights. These insights explain the significance of important financial indicators such as profitability, return ratios, cash flow performance, valuation, and overall financial health in a clear and understandable manner.

By converting quantitative information into qualitative summaries, the module makes the platform more accessible to users with different levels of financial knowledge. Investors can quickly understand the strengths and weaknesses of a company, while students and researchers can use the generated insights to better interpret financial statements and analytical results.

Key Features

- Rule-based generation of financial insights.
- Automatic interpretation of key financial metrics.
- Company-specific business summaries.
- Easy-to-read textual analysis alongside interactive dashboards.
- Integration with Financial Health Score and profitability indicators.
- Dynamic updates based on user-selected company.

Business Value

Financial dashboards often present a large amount of numerical information, which can be difficult to interpret without financial expertise. The NLP Insights module bridges this gap by converting analytical outputs into business-oriented explanations. This improves decision-making efficiency and reduces the time required to understand complex financial data.

The module also supports educational use by explaining financial performance in plain language, making it valuable for students learning financial analysis. For investors and analysts, the generated insights provide a quick overview before conducting more detailed investigations.

Overall, the NLP Insights module combines financial analytics with natural language generation to create a more intuitive and user-friendly analytical experience within the N100 Financial Intelligence Platform.

Figure 20.1: NLP Insights Dashboard of the N100 Financial Intelligence Platform.

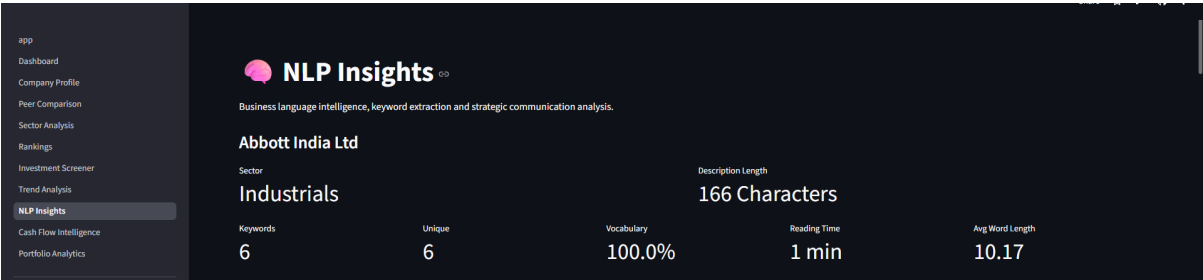


Figure 20.1: Application screenshot supporting this module.

Cash Flow Intelligence

The Cash Flow Intelligence module provides a comprehensive evaluation of how effectively a company generates, utilizes, and manages cash. While profitability metrics indicate whether a company is earning profits, cash flow analysis reveals whether those profits are supported by actual cash generation. For this reason, cash flow is considered one of the most important indicators of long-term financial stability and business sustainability.

This module consolidates key cash flow metrics into an interactive dashboard that enables users to evaluate the quality of a company’s operations. Users can analyze Operating Cash Flow, Free Cash Flow, Capital Expenditure (CapEx), and other liquidity-related indicators through dynamic visualizations and KPI cards. These metrics help determine whether a business generates sufficient cash to fund operations, invest in future growth, and meet its financial obligations.

The dashboard presents financial information in an intuitive format using interactive charts and summary cards. Users can quickly compare cash flow performance across companies, identify organizations with strong liquidity, and recognize businesses that may face operational or financing challenges.

Key Features

- Operating Cash Flow analysis.
- Free Cash Flow evaluation.
- Capital Expenditure (CapEx) insights.
- Interactive KPI cards.
- Company-level cash flow visualization.
- Dynamic charts for financial interpretation.
- Integration with Financial Health Score.

Business Value

Cash flow analysis complements traditional profitability analysis by focusing on the actual movement of cash rather than accounting earnings. A company with strong cash flow is generally better positioned to invest in expansion, repay debt, distribute dividends, and withstand economic uncertainty.

The Cash Flow Intelligence module provides investors, analysts, and researchers with valuable insights into a company’s liquidity and financial resilience. By combining multiple cash flow indicators within a single dashboard, the platform enables faster, more informed decision-making and supports comprehensive financial analysis.

Overall, this module strengthens the analytical capabilities of the N100 Financial Intelligence Platform by highlighting the financial sustainability of companies beyond conventional income statement metrics.

Figure 21.1: Cash Flow Intelligence Dashboard - Overview.

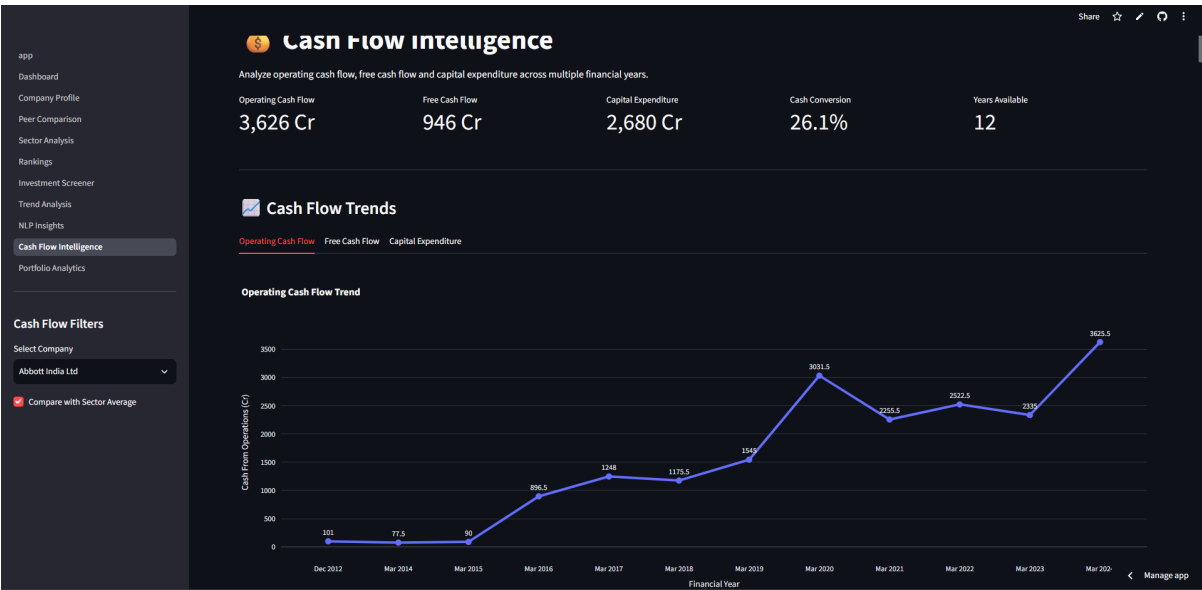


Figure 21.1: Application screenshot supporting this module.

Cash Flow Analysis and Financial Interpretation

The second section of the Cash Flow Intelligence module focuses on interpreting the quality of a company's cash generation and its ability to sustain long-term operations. While the overview dashboard presents headline cash flow metrics, this section provides deeper analytical insights by comparing operating cash flow, free cash flow, capital expenditure, and other liquidity indicators through interactive visualizations.

A company's cash flow reflects its operational efficiency more accurately than accounting profits because it measures actual cash moving into and out of the business. Positive and consistently growing operating cash flow generally indicates that the company's core business operations are healthy and capable of supporting future expansion without excessive dependence on external financing.

The dashboard allows users to compare cash flow performance across multiple companies and sectors. Interactive charts make it easier to identify organizations that efficiently convert revenue into cash while highlighting businesses that may experience liquidity constraints despite reporting accounting profits.

Key Financial Metrics

- Operating Cash Flow (OCF)
- Free Cash Flow (FCF)
- Capital Expenditure (CapEx)
- Cash Flow Efficiency
- Cash Generation Trends
- Company-wise Cash Flow Comparison

Business Insights

The visualizations presented in this section help users answer important financial questions, such as:

- Is the company generating sufficient operating cash?
- Does the company consistently produce positive free cash flow?
- How much cash is being reinvested into business operations?
- Which companies demonstrate stronger liquidity compared to their peers?
- Are current cash flow patterns sustainable over the long term?

These insights enable investors and analysts to evaluate a company's financial resilience beyond traditional profitability measures and make better-informed investment decisions.

Overall, this section complements the financial ratios available in other modules by providing a detailed understanding of cash movement, liquidity, and operational sustainability.

Figure 22.1: Cash Flow Intelligence - Company Cash Flow Analysis.



Figure 22.1: Application screenshot supporting this module.

Advanced Cash Flow Visual Analytics

The third section of the Cash Flow Intelligence module focuses on advanced visual exploration of cash flow performance. While the previous sections emphasize key metrics and company-level interpretation, this page highlights comparative visualizations that allow users to identify patterns, distributions, and relationships across multiple companies and sectors.

Interactive charts such as scatter plots, treemaps, distribution charts, and comparative visualizations enable users to understand how cash flow performance varies across the Nifty 100 ecosystem. These visual tools simplify the identification of financially strong organizations, companies with inefficient cash utilization, and sector-specific cash flow trends.

One of the major strengths of this module is its ability to present complex financial relationships in a graphical format. Users can quickly recognize outliers, clusters of high-performing companies, and differences in liquidity management without manually reviewing financial statements.

Key Visual Components

- Interactive bubble charts for company comparison.
- Distribution analysis of cash flow metrics.
- Treemap visualization of sector-wise cash flow.
- Comparative charts showing Operating Cash Flow and Free Cash Flow.
- Financial Health Score relationship with cash flow performance.
- Interactive filtering by company and sector.

Business Insights

The advanced visualizations support several important analytical objectives:

- Identify companies generating consistently strong cash flows.
- Compare liquidity performance across different industries.
- Detect organizations with unusually high or low capital expenditure.
- Recognize sector-level patterns in cash generation.
- Evaluate the relationship between cash flow quality and overall financial health.

These graphical insights make financial analysis significantly faster and more intuitive than relying solely on numerical tables. They also improve communication of business findings to investors, managers, and stakeholders.

Overall, this section strengthens the analytical capability of the N100 Financial Intelligence Platform by combining financial intelligence with modern data visualization techniques, enabling users to uncover trends and relationships that may not be immediately apparent from raw financial data.

Figure 23.1: Advanced Cash Flow Visual Analytics Dashboard.

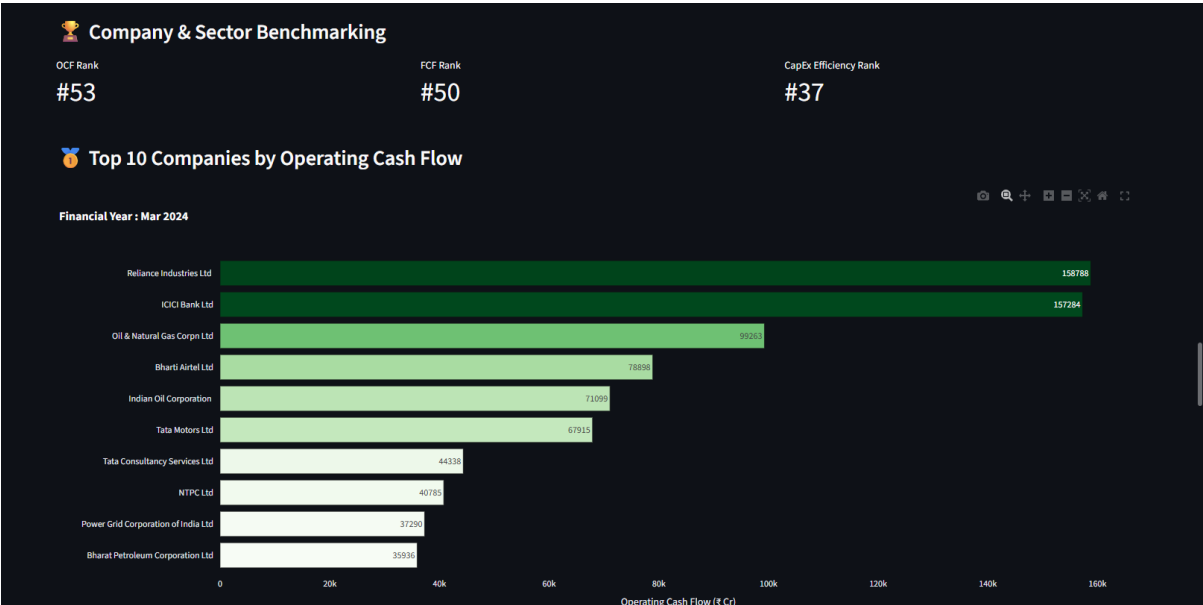


Figure 23.1: Application screenshot supporting this module.

Cash Flow Benchmarking and Business Insights

The final section of the Cash Flow Intelligence module focuses on benchmarking companies using cash flow-related indicators and summarizing the financial insights obtained from previous analyses. After evaluating individual companies and visualizing cash flow trends, this section enables users to compare organizations against industry peers and identify businesses with superior financial sustainability.

Benchmarking is an important step in financial analysis because a company's performance should always be interpreted relative to comparable organizations. The dashboard compares companies using metrics such as Operating Cash Flow, Free Cash Flow, Capital Expenditure, Cash Flow Efficiency, and the Financial Health Score. These comparisons help users distinguish financially resilient companies from those experiencing weaker liquidity or inefficient cash utilization.

The module also highlights sector-level averages and identifies companies that consistently outperform their peers. Through interactive charts and summary tables, users can evaluate whether a company's cash generation aligns with broader industry trends and assess its long-term financial stability.

Key Features

- Company-wise cash flow benchmarking.
- Sector average comparison.
- Financial Health Score benchmarking.
- Interactive KPI cards and summary tables.
- Company ranking based on cash flow performance.
- Comparative charts for identifying industry leaders.
- Business insight summaries supporting investment decisions.

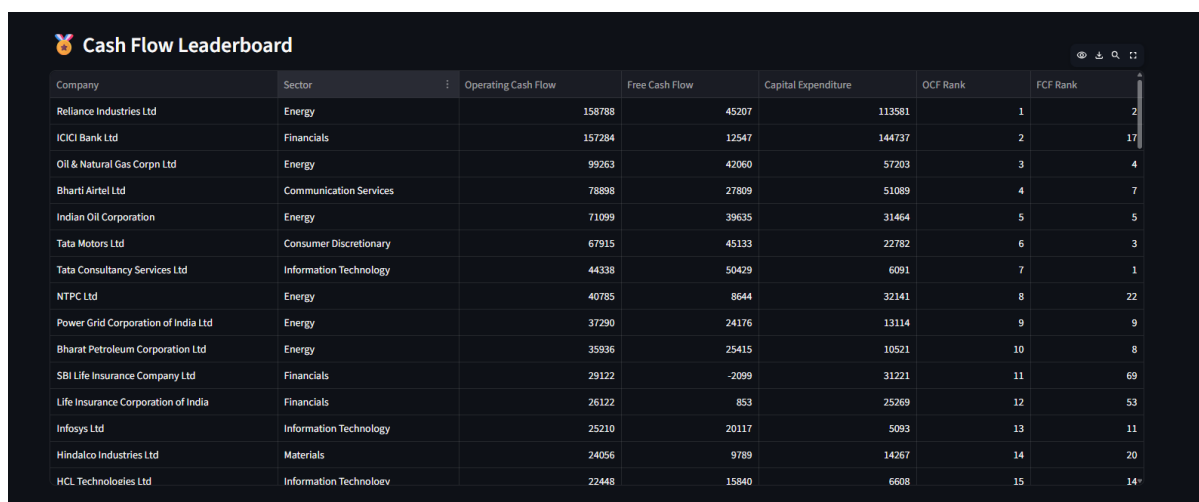
Business Value

Benchmarking provides context to financial analysis by helping users understand whether a company's cash flow performance is exceptional, average, or below industry expectations. Investors can identify financially strong businesses with sustainable cash generation, while analysts can compare liquidity performance across sectors to support strategic recommendations.

The insights generated in this section complement profitability analysis by emphasizing cash-based performance, enabling more balanced investment decisions. Together with the previous Cash Flow Intelligence pages, this module offers a comprehensive framework for evaluating operational efficiency, financial resilience, and long-term sustainability.

Overall, the Cash Flow Intelligence module demonstrates how interactive analytics can transform complex financial information into actionable business intelligence for investors, researchers, students, and decision-makers.

Figure 24.1: Cash Flow Benchmarking Dashboard of the N100 Financial Intelligence Platform.



Cash Flow Leaderboard

Company	Sector	Operating Cash Flow	Free Cash Flow	Capital Expenditure	OCF Rank	FCF Rank
Reliance Industries Ltd	Energy	158788	45207	113581	1	2
ICICI Bank Ltd	Financials	157284	12547	144737	2	17
Oil & Natural Gas Corpn Ltd	Energy	99263	42060	57203	3	4
Bharti Airtel Ltd	Communication Services	78898	27809	51089	4	7
Indian Oil Corporation	Energy	71099	39635	31464	5	5
Tata Motors Ltd	Consumer Discretionary	67915	45133	22782	6	3
Tata Consultancy Services Ltd	Information Technology	44338	50429	6091	7	1
NTPC Ltd	Energy	40785	8644	32141	8	22
Power Grid Corporation of India Ltd	Energy	37290	24176	13114	9	9
Bharat Petroleum Corporation Ltd	Energy	35936	25415	10521	10	8
SBI Life Insurance Company Ltd	Financials	29122	-2099	31221	11	69
Life Insurance Corporation of India	Financials	26122	853	25269	12	53
Infosys Ltd	Information Technology	25210	20117	5093	13	11
Hindalco Industries Ltd	Materials	24056	9789	14267	14	20
HCL Technologies Ltd	Information Technology	22448	15840	6608	15	14*

Figure 24.1: Application screenshot supporting this module.

Portfolio Analytics

The Portfolio Analytics module enables users to build and evaluate a portfolio consisting of multiple Nifty 100 companies within a single interactive environment. Rather than analyzing companies individually, this module focuses on how a collection of investments performs collectively by providing insights into diversification, sector allocation, financial strength, and comparative performance.

Users can select multiple companies to construct a custom portfolio and immediately visualize aggregated financial metrics using interactive dashboards. The module combines portfolio-level KPIs, financial summaries, charts, and comparison tables to provide a comprehensive overview of the selected investments.

The dashboard is designed to help users understand the composition of their portfolio, evaluate company-wise contributions, and identify strengths or weaknesses in their investment allocation. By integrating multiple financial indicators into one interface, the platform supports data-driven portfolio evaluation and encourages balanced investment decisions.

Key Features

- Interactive portfolio builder.
- Selection of multiple Nifty 100 companies.
- Portfolio-level KPI cards.
- Company-wise financial summaries.
- Sector allocation analysis.
- Interactive charts and tables.
- Integration with Financial Health Score.

Business Value

A diversified portfolio reduces investment risk by spreading capital across different companies and sectors. The Portfolio Analytics module assists users in evaluating whether their selected companies are well balanced and financially sound. It also highlights concentration risks by displaying sector distributions and comparative performance indicators.

This module is particularly valuable for investors who wish to compare multiple companies before allocating capital. Students and researchers can also use it to study portfolio construction techniques and understand how diversification affects overall financial performance.

Overall, the Portfolio Analytics module transforms individual company analysis into a holistic portfolio-level assessment, enabling users to make more informed investment decisions using interactive business intelligence tools.

Figure 25.1: Portfolio Analytics Dashboard - Portfolio Overview.

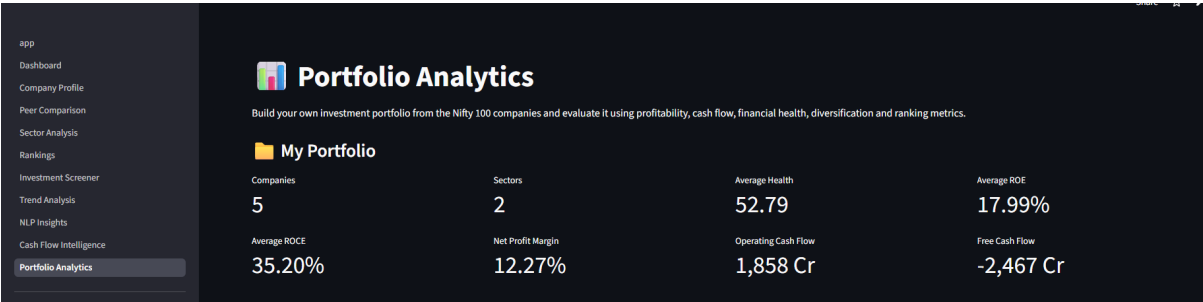


Figure 25.1: Application screenshot supporting this module.

Portfolio Builder and Performance Analysis

The second section of the Portfolio Analytics module focuses on portfolio construction and performance evaluation. After selecting multiple companies, users can examine how the portfolio is distributed across sectors and compare the financial performance of each constituent company. This interactive analysis provides a comprehensive understanding of portfolio composition and highlights opportunities for improving diversification.

The Portfolio Builder allows users to create a customized portfolio by selecting companies from the Nifty 100 index. As companies are added or removed, the dashboard dynamically updates all key performance indicators, charts, and summary tables. This real-time interaction enables users to immediately observe the impact of portfolio composition on overall financial performance.

Performance analysis is presented using a combination of KPI cards, sector allocation charts, comparison tables, and interactive visualizations. These components summarize portfolio characteristics such as the number of selected companies, average Financial Health Score, profitability metrics, and sector distribution. Users can easily identify whether the portfolio is concentrated in a single industry or diversified across multiple sectors.

Key Features

- Interactive portfolio builder with company selection.
- Dynamic portfolio performance summary.
- Sector allocation visualization.
- Average Financial Health Score of selected companies.
- Company-wise comparison table.
- Interactive performance charts.
- Automatic updates based on user selections.

Business Value

Portfolio construction is one of the most important aspects of investment management. A well-diversified portfolio helps reduce overall investment risk while improving long-term stability. This module enables users to evaluate whether their selected companies provide sufficient diversification and balanced exposure across industries.

The visualizations simplify portfolio evaluation by presenting complex financial information in an intuitive format. Investors can compare companies, identify overexposure to particular sectors, and make more informed allocation decisions. Students and researchers can also use the module to understand the practical application of diversification principles in portfolio management.

Overall, the Portfolio Builder and Performance Analysis section demonstrates how interactive business intelligence tools can support effective portfolio construction and financial decision-making.

Figure 26.1: Portfolio Builder and Performance Analysis Dashboard.

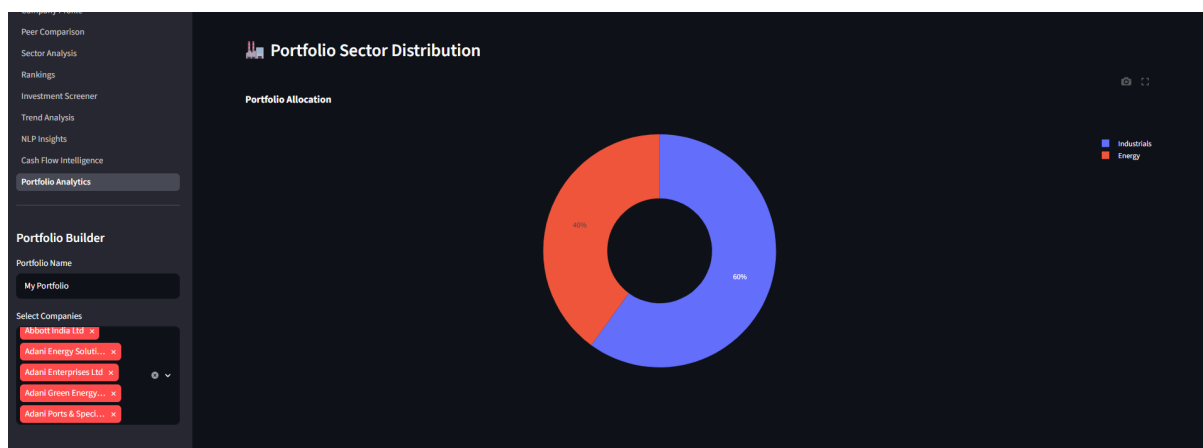


Figure 26.1: Application screenshot supporting this module.

Financial Analysis and Comparative Insights

The third section of the Portfolio Analytics module focuses on evaluating the financial quality of the selected portfolio by comparing companies across multiple financial dimensions. While the previous section emphasized portfolio construction and sector allocation, this page provides a deeper analysis of individual company performance using profitability metrics, return ratios, cash flow indicators, and the Financial Health Score.

The dashboard combines interactive charts, comparison tables, and statistical visualizations to help users understand how each company contributes to the overall portfolio. Metrics such as Return on Equity (ROE), Return on Capital Employed (ROCE), profitability, and cash flow performance are displayed side by side, allowing users to identify both strong performers and companies that may require further evaluation.

Interactive visualizations, including box plots, bar charts, and comparative financial graphs, highlight differences between companies and reveal the distribution of important financial metrics within the portfolio. These visual tools simplify complex financial analysis and allow users to quickly recognize patterns, outliers, and inconsistencies.

Key Features

- Comparison of Financial Health Scores across portfolio companies.
- ROE and ROCE comparison charts.
- Cash flow performance analysis.
- Profitability comparison across selected companies.
- Statistical distribution of financial metrics using box plots.
- Interactive charts for deeper financial evaluation.
- Company-level benchmarking within the portfolio.

Business Value

A portfolio should not only be diversified but also consist of financially strong companies. This module enables users to evaluate the quality of each investment by comparing standardized financial indicators within the portfolio. Investors can identify organizations with consistently strong profitability and cash generation while recognizing weaker performers that may increase portfolio risk.

The comparative analysis also supports researchers and students by demonstrating how multiple financial metrics can be integrated to evaluate investment quality. Interactive visualizations improve the interpretability of financial information and encourage evidence-based decision-making.

Overall, this section enhances the Portfolio Analytics module by providing detailed financial comparisons that complement portfolio-level summaries and strengthen the overall investment analysis process.

Figure 27.1: Portfolio Financial Analysis and Comparative Insights Dashboard.

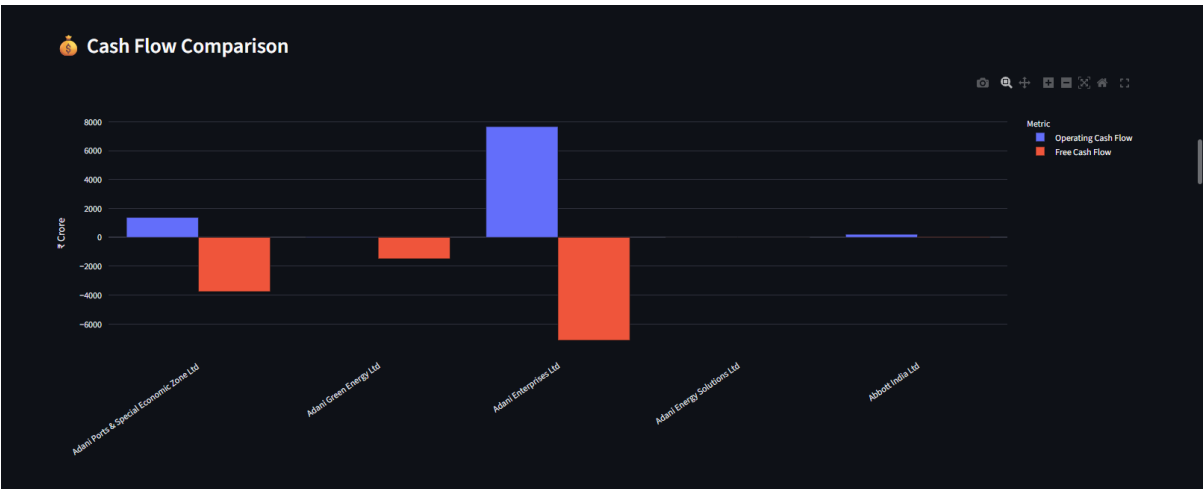


Figure 27.1: Application screenshot supporting this module.

Portfolio Leaderboard and Strategic Positioning

The final section of the Portfolio Analytics module presents a comprehensive ranking and benchmarking of the companies included in the selected portfolio. While the previous sections focused on portfolio construction, sector allocation, and financial comparisons, this page highlights the relative position of each company using leaderboards, benchmarking tables, and strategic visualizations.

The dashboard ranks portfolio companies using important financial indicators such as the Financial Health Score, profitability ratios, return metrics, and cash flow performance. By consolidating these metrics into an easy-to-understand leaderboard, users can quickly identify the strongest and weakest performers within their portfolio. This ranking provides valuable support for portfolio review, rebalancing, and long-term investment planning.

In addition to rankings, the dashboard includes interactive visualizations such as portfolio positioning charts, sector allocation summaries, treemaps, and comparative graphs. These visuals help users understand whether the portfolio is well diversified, identify companies contributing the most to overall quality, and recognize areas where additional diversification may be beneficial.

Key Features

- Portfolio leaderboard ranked by Financial Health Score.
- Top-performing and bottom-performing company summaries.
- Portfolio positioning through interactive bubble charts.
- Sector allocation and diversification analysis.
- Comparative benchmarking across selected companies.
- Treemap visualization of portfolio composition.
- Interactive filtering and dynamic portfolio updates.

Business Value

Portfolio evaluation is an ongoing process that requires regular monitoring of investment quality and diversification. The leaderboard simplifies this process by presenting financial performance in a ranked format, enabling users to identify companies that consistently outperform or underperform the rest of the portfolio.

The strategic positioning visualizations provide additional context by showing how companies are distributed across sectors and financial performance categories. Investors can use these insights to rebalance portfolios, reduce concentration risk, and improve long-term investment decisions. Researchers and students can also use this module to understand practical portfolio evaluation techniques supported by modern business intelligence tools.

Overall, the Portfolio Leaderboard and Strategic Positioning section completes the Portfolio Analytics module by transforming financial comparisons into actionable investment insights, making it one of the most comprehensive analytical components of the N100 Financial Intelligence Platform.

Figure 28.1: Portfolio Leaderboard and Strategic Positioning Dashboard.



Figure 28.1: Application screenshot supporting this module.

Project Deployment

After the successful development and testing of the N100 Financial Intelligence Platform, the application was deployed as a publicly accessible web application using Streamlit Community Cloud. Deployment transforms the project from a local development environment into a production-ready application that can be accessed through any modern web browser without requiring users to install software or configure dependencies.

The deployment process began with organizing the project into a modular directory structure containing the application source code, processed datasets, database files, configuration files, documentation, and project assets. Version control was maintained using Git, and the complete project was hosted on GitHub, enabling efficient source code management, collaboration, and continuous updates.

The GitHub repository was then connected to Streamlit Community Cloud, which automatically built and deployed the application. During deployment, the required Python packages were installed from the project's dependency file, and the Streamlit application was configured to launch the main dashboard. This cloud-based deployment allows users to explore all analytical modules through a responsive web interface from any location.

Deployment Workflow

1. Develop and test the application locally using Streamlit.
2. Organize the project into a clean repository structure.
3. Push the complete project to GitHub.
4. Connect the GitHub repository to Streamlit Community Cloud.
5. Install project dependencies automatically.
6. Deploy the application and verify all dashboard modules.
7. Share the public application link for user access.

Deployment Outcomes

- Publicly accessible web application.
- Version-controlled source code hosted on GitHub.
- Interactive dashboards available online.
- Cross-platform accessibility through web browsers.
- Simplified maintenance and future updates.

Project Links

GitHub Repository

https://github.com/Bhardwaj-Arin/N100_FINANCIAL_INTELLIGENCE_PLATFORM

Live Streamlit Application

<https://n100financialintelligenceplatform-njqenrfgue6evxbdjcmj.streamlit.app/>

Figure 29.1: GitHub Repository and Deployed Streamlit Application.

Testing and Validation

Testing and validation were carried out throughout the development of the N100 Financial Intelligence Platform to ensure that every analytical module produced accurate, reliable, and consistent results. Since the platform integrates data processing, financial calculations, database operations, and interactive dashboards, each stage of the workflow was verified before deployment.

The testing process began with validating the raw financial dataset. Missing values, duplicate records, inconsistent formats, and incorrect data types were identified and corrected during preprocessing. Data validation checks ensured that only clean and consistent information was passed to the feature engineering stage.

Financial calculations such as profitability ratios, return metrics, cash flow indicators, and the Financial Health Score were verified against the processed dataset to confirm that the implemented formulas generated accurate values. Database testing confirmed that records were stored and retrieved correctly from the SQLite database without data loss or inconsistencies.

Each Streamlit module was manually tested to verify navigation, filters, charts, tables, and user interactions. Plotly visualizations were checked for correctness, responsiveness, and consistency across different company selections. The deployed application was also tested on multiple web browsers to ensure that all dashboards loaded correctly and interacted as expected.

Testing Activities

- Validation of raw financial data.
- Data cleaning verification.
- Feature engineering validation.
- Financial metric verification.
- SQLite database testing.
- Dashboard functionality testing.
- Interactive filter validation.
- Chart and visualization verification.
- End-to-end application testing.
- Deployment verification on Streamlit Community Cloud.

Test Results

The completed testing process confirmed that all implemented modules function correctly and produce consistent analytical outputs. Interactive dashboards respond accurately to user inputs, financial metrics are calculated correctly, and visualizations display reliable business insights. The deployed application successfully delivers a stable and user-friendly experience suitable for demonstration, academic submission, and portfolio presentation.

Figure 30.1: Testing and Validation Workflow of the N100 Financial Intelligence Platform.

Results and Achievements

The successful development of the N100 Financial Intelligence Platform demonstrates the practical application of data engineering, financial analytics, business intelligence, database management, and interactive dashboard development within a single end-to-end solution. The completed platform transforms raw financial information into meaningful insights that support informed decision-making for investors, analysts, researchers, and students.

The project achieved its primary objective of creating a centralized financial intelligence platform for companies listed in the Nifty 100 index. Through systematic data processing, feature engineering, financial metric computation, and interactive visualization, the application provides users with an intuitive environment for exploring company performance and market trends.

Major Achievements

- Successfully developed a production-style financial analytics platform.
- Implemented 10 interactive analytical modules, each addressing a different aspect of financial analysis.
- Built a complete data processing pipeline including data validation, cleaning, feature engineering, and database integration.
- Calculated important financial indicators such as profitability ratios, return metrics, cash flow measures, and a composite Financial Health Score.
- Developed interactive dashboards using Streamlit and Plotly for real-time exploration of financial data.
- Enabled company profiling, peer comparison, sector benchmarking, investment screening, trend analysis, NLP-based financial summaries, cash flow intelligence, and portfolio analytics.
- Integrated a SQLite database for efficient storage and retrieval of processed financial information.
- Successfully deployed the application using Streamlit Community Cloud and maintained the source code using GitHub.

Learning Outcomes

This project significantly strengthened practical skills in:

- Python programming
- Data preprocessing and feature engineering
- Financial statement analysis
- Business intelligence and dashboard development
- Database management using SQLite
- Interactive data visualization with Plotly
- Web application development using Streamlit
- Version control using Git and GitHub
- Software testing and deployment

Overall, the N100 Financial Intelligence Platform demonstrates how modern data science techniques can be combined with financial analytics to build a scalable, user-friendly decision-support system capable of transforming complex financial data into actionable business intelligence.

Figure 31.1: Summary of Major Project Achievements.

Conclusion

The N100 Financial Intelligence Platform successfully achieves its objective of transforming raw financial data into an interactive, comprehensive, and user-friendly decision support system. By integrating data preprocessing, feature engineering, financial analysis, database management, and business intelligence dashboards into a single platform, the project demonstrates a complete end-to-end data science workflow.

The platform provides meaningful insights for all companies in the Nifty 100 index through modules such as Company Profile, Peer Comparison, Sector Analysis, Rankings, Investment Screener, Trend Analysis, NLP Insights, Cash Flow Intelligence, and Portfolio Analytics. Each module contributes a unique perspective, enabling users to evaluate companies from multiple financial dimensions rather than relying on a single metric.

From a technical perspective, the project showcases the practical application of Python, Pandas, SQLite, Plotly, and Streamlit to build a scalable analytics solution. The modular architecture makes the system easy to maintain, extend, and deploy while ensuring that analytical computations remain accurate and consistent across all dashboard components.

Beyond fulfilling the project requirements, the platform serves as a strong demonstration of modern data engineering, financial analytics, and dashboard development skills. It illustrates how data science can be applied to solve real-world business problems by converting complex financial information into actionable insights for investors, analysts, researchers, and students.

Future Scope

Although the current platform provides extensive financial intelligence capabilities, several enhancements can further improve its functionality and real-world applicability.

Potential Enhancements

- Integrate real-time stock market and financial statement data through APIs.
- Add machine learning models for stock trend forecasting and financial risk prediction.
- Implement personalized investment recommendations based on user preferences.
- Introduce portfolio optimization using modern portfolio theory and risk-return analysis.
- Develop automated financial report generation in PDF format.
- Add user authentication, profiles, and cloud-based portfolio management.
- Expand the platform to include global stock markets beyond the Nifty 100.
- Deploy the application on scalable cloud infrastructure with continuous integration and deployment (CI/CD).

By incorporating these enhancements, the platform can evolve from an academic project into a production-ready financial intelligence solution capable of supporting a wider range of users and business applications.

Figure 32.1: Future Roadmap for the N100 Financial Intelligence Platform.

Future Scope and Enhancement Roadmap

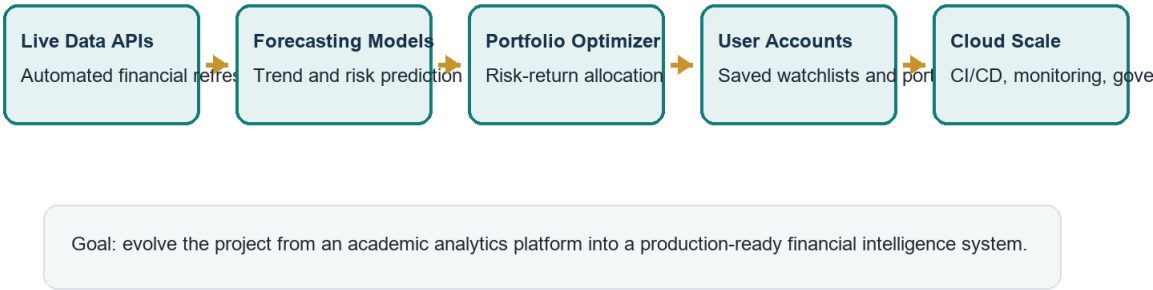


Figure 32.1: Future Scope and Enhancement Roadmap for the N100 Financial Intelligence Platform.

References and Repository Assets

Reference / Asset	Description
GitHub Repository	https://github.com/Bhardwaj-Arin/N100_FINANCIAL_INTELLIGENCE_PLATFORM
Live Streamlit Application	https://n100financialintelligenceplatform-njqenrfxgue6evxbdjcmj.streamlit.app/
data/raw Excel files	Raw company, ratio, market, sector, stock price, balance sheet, profit and loss, and cash-flow data.
data/processed/master_features.csv	Final processed analytical dataset for scoring and dashboard modules.
SQLite database files	Local databases used to store company features and financial health score outputs.
Official documentation	Python, Pandas, NumPy, Plotly, Streamlit, SQLite, GitHub, and Streamlit Community Cloud.